Chapter 3

Control of Microorganisms by Chemical Methods

Learning Objectives

After studying this chapter the students will be able to,

- *Defines the terms disinfectants, antiseptics and antibiotics*
- *Describe major groups of antimicrobial chemical agents and uses of disinfectants.*
- *Describe the factors related to effective disinfectants.*
- *Discuss the classification of antibiotics and their mode of action.*
- *Know the procedure used in antimicrobial susceptibility testing in clinical laboratory.*
- *Knows the resistance mechanisms developed by pathogens against antibiotic or chemotherapy drugs.*

Chapter Outline

3.1 Disinfectants, Antiseptics and Antibiotics

3.2 Factors Influencing the Antimicrobial Activity of Chemical Agents

3.3 Mode of Action of Chemical Agents

3.4 Major Groups of Antimicrobial Chemical Agents

3.5 Evaluation of Antimicrobial Chemical Agents

3.6 Antibiotics

3.7 Antimicrobial Susceptibility Testing

3.8 Drugs Resistance Mechanisms

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Control of microorganisms is essential in order to prevent the transmission of diseases, infection, spoilage and to or remove unwanted microbial contamination. Microorganisms are controlled by means of physical agents and chemical agent. In 11th standard, we learnt different physical methods of sterilization. Control by chemical agents refers to the use of disinfectants, antiseptics, antibiotics and chemotherapeutic antimicrobial chemicals. This chapter describes various chemical agents, their mode of action, and their evaluation.

Use of chemicals to sterilize objects and to control microbial pathogen from causing diseases has been in practice since centuries. A large number of chemicals are now available for this purpose. Commercial products which incorporate these chemicals are used in a variety of conditions and they usually differ in their mode of action. No single chemical agent is best for any and all purposes. Hence several classes of chemicals have been identified and new compounds are developed that possess destructive properties in terms of their suitability for practical application.

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